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Mango leaf disease detection and classification using a federated learning based lightweight vision transformer model

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Abstract: The detection and management of horticultural crop diseases is crucial for enhancing productivity, reducing costs, and promoting environmentally sustainable practices. Recently, various techniques have been developed for managing crop-related issues, including leaf diseases and pest control. A centralized control approach has been implemented for disease detection. This centralized system faced significant challenges, including concerns over data privacy and the costs associated with data migration. This study addressed the problems with centralized solutions through a federated learning approach enhanced by the attention mechanism of the vision transformer (ViT). We investigated the key features of the federated learning method and propose a deep learning model based on federated learning, utilizing the attention mechanism of the ViT to detect diseases in mango plant leaves. The study utilized an open-access dataset of mango leaf diseases sourced from Kaggle. The performance of the federated model depended heavily on several factors, including data quality, the number of participating agents, their communication efficiency, and the number of local iterations. Although various pretrained models were tested to address these challenges, the lightweight ViT was the most effective, achieving an impressive accuracy rate of 98.99% in federated learning scenarios.

Key words: Deep learning, federated learning, crop leaf disease, horticulture

1. Introduction

Data science techniques have been used for plant leaf disease detection and management, and models have been developed for low-cost, efficient, and environmentally sustainable solutions to improve crop productivity (Kaur et al., 2022; Khullar et al., 2025). Precision agriculture relying on modern technologies improves crop productivity. Various revolutionized modern terminologies have been used under precision agriculture to enhance productivity (Geetharamani and Arun Pandian, 2019). Better disease detection solutions reduce the dependency on pesticides and plant protection products (Li et al., 2021; Ouhami et al., 2021, Antico et al., 2024). This ultimately impacts the economy, crop quality, and yield. In recent years, various technologies have been used to develop sustainable solutions for plant disease detection (Tugrul et al., 2022). Deep neural networks are widely used for resolving agriculture issues (Borhani et al., 2022). Despite these techniques yielding excellent results and surpassing expectations, they encounter challenges concerning data

security due to the storage of data on a centralized server. Thus, additional development is needed to address security concerns for centrally stored data.

Federated learning (FL) is the most promising approach to resolve issues with centralized approaches. Centrally stored data is used to train artificial intelligence models and uncover valuable information on production techniques (Maillet et al., 2023) and disease management. However, accessing this data brings with it the risk of data breaches. FL enables entities to collaborate or work together without sharing their private data and maintain confidentiality and data security. FL empowers the model both remotely and locally. Moreover, it minimizes data sharing and provides full data control to the user and minimizes data leakage (Kerkech et al., 2020). FL ensures privacy policy adherence and alleviates resource utilization. All the participants in FL benefit from the model trained on quality data without the security risks (Wen et al., 2023). Although FL is a very effective approach, there are various challenges such complexity of

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coordination and data heterogeneity. In this paper, various pretrained models were evaluated for the characterization of mango leaf disease in an FL environment. The models were evaluated on a public domain dataset. The objective of this paper was to highlight the benefits of FL in mango leaf disease detection. Using FL, the proposed technique was incorporated with data security and confidentiality of sensitive data. The goals of this study were as follows:

1. To investigate the essential elements of the FL approach to identify diseases in mango plant leaves. We propose a deep learning model that leverages the attention mechanism of the vision transformer (ViT) to enhance FL.
2. Investigate the effects of the attention mechanism on identified plant leaf diseases utilizing an FL convolutional neural networks (CNN) model.
3. Examine the influence of different devices and the number of iterations within those devices in the leaf disease detection experiment. Additionally, the benefits of FL for detecting plant leaf diseases were explored.

The state of the art in the field is presented in Section 2. The proposed methodology with implementation details is presented in Section 3. The results and discussion are presented in Section 4. Finally, the conclusion of the article is presented in Section 5.

2. Related work

This section summarizes centralized techniques for leaf disease detection. Several state-of-the-art models are available that process data jointly or individually.

Despite the increasing interest in the implementation of FL across various fields, its utilization in the agricultural industry, especially for the detection of plant leaf disease, remains limited. A review of the current literature indicates that only a handful of studies have addressed important issues such as data privacy, security, and disease detection in agriculture. Many challenges have not been addressed, including the analysis of disease severity, weed detection, soil fertility assessment, and pest identification (Abdulridha et al., 2016; Ganatra and Patel, 2020; Xu et al., 2021).

This paper by Chimate et al. (2025) examines effective feature extraction methods designed to enhance the accuracy of disease classification in mango and groundnut crops. By refining feature extraction techniques, the study improved the performance of plant disease detection systems, which could lead to better management of crop health. They emphasize advanced feature extraction strategies to bolster disease identification processes in agricultural engineering. The paper describes a method for detecting pests that affect mangoes through advanced statistical and machine learning techniques—the Gaussian mixture model and expectation-maximization

algorithm—for more accurate detection (Pansy, 2025). Swapno et al. (2024) developed a CNN and generated encouraging results, with accuracy rates between 93.12% and 96.67% in identifying various mango leaf diseases. The innovative deep learning framework utilizes ensemble and transfer learning techniques to accurately identify mango leaf diseases, achieving an impressive 94.99% testing accuracy.

Neog and Kaur (2025) integrated multiple neural network models (ResNet50, VGG19, MobileNet, and a custom CNN) to improve detection efficiency and robustness in mango leaf disease classification, exceeding the performance of individual models in critical metrics like precision and recall. The system uses a comprehensive dataset of 8000 images, data augmentation, and advanced training strategies to enable early disease diagnosis, thus supporting sustainable mango cultivation and crop management (Neog and Kaur, 2025). The study by Varma et al. (2025) investigated the application of various transfer learning models to automatically identify diseases in mango leaves, highlighting the significance of early and precise diagnosis for effective crop management. The research by Varma et al. (2025) introduces a deep learning methodology that uses the EfficientNetB5 model to effectively categorize mango leaf images into eight distinct groups, encompassing six types of diseases, healthy leaves, and dieback, achieving a remarkable test accuracy of 97%.

Aluvala and Chauhan (2025) suggested a framework that serves as a promising tool for accurate disease diagnosis, contributing to sustainable agriculture, food security, and economic growth, with the potential for real-time application in agricultural practices. They outline an innovative strategy that merges deep learning with ensemble learning techniques to enhance the diagnosis of mango leaf diseases. By utilizing advanced machine learning models, the proposed approaches seeks to improve accuracy and reliability in disease detection, which is essential for timely intervention and effective crop management. These comprehensive methods address the shortcomings of conventional diagnostic techniques and provide a valuable resource for agricultural decision-making (Gautam et al., 2024; Gautam et al., 2025).

This article introduced a groundbreaking method for detecting mango diseases by integrating a fused ViT with ConvNeXt architecture, with the goal of improving accuracy and robustness in disease identification. Combining sophisticated deep learning models to analyze intricate visual patterns in mango leaf images ultimately promotes better crop health management (Alamri et al., 2025). Alamri et al. (2025) investigated the impact of preprocessing techniques on classification accuracy using five CNN models, both prior to and following data preprocessing. The results indicate significant improvements in accuracy

after preprocessing, emphasizing its vital role in enhancing model performance. Additionally, a novel custom model is proposed and assessed, showcasing its capability to achieve an impressive accuracy of 98.50% (Das et al., 2025).

A streamlined parallel multiscale attention CNN (PMACNN) architecture has been created to minimize communication expense. Evaluation of simulated federated datasets attains an impressive testing accuracy of 97.5% (Hari and Singh, 2025). Data related to various agriculture domains such as weed management, disease management, and soil management are distributed to various devices. Table 1 summarizes the state of the art of leaf disease detection with centralized and federated approaches.

In centralized machine learning systems, all client-side data must be sent to a central server for model training and inference. Although centralized methods—such as those described by Zhong and Zhao (2020) and Mac et al. (2024)—perform well when utilizing architectures like

DenseNet121 and VGG19, they raise concerns in real-world agriculture. Farmers, agritech companies, and cooperatives frequently function in data-sensitive environments, where insights regarding crop conditions, disease trends, and yield predictions are viewed as confidential or proprietary. Research indicates the reluctance of stakeholders to share unprocessed agricultural data due to concerns over privacy violations, market manipulation, and data exploitation.

Moreover, remote farm locations with limited data connectivity increases the cost and latency challenges with large-scale data transfer, rendering centralized methods impractical on a larger scale. These constraints have propelled recent investigations—such as those conducted by Vats et al. (2024), Singh et al. (2024), and Hari and Singh (2025)—toward FL. FL is a decentralized framework that facilitates collaborative model training without the need for raw data exchange. The growing use of FL signifies a paradigm shift aimed at achieving both model precision and data sovereignty in precision agriculture (Table 1).

Table 1. Techniques to recognize leaf disease.

Research	Year	Leaf type	Technique	Approach	Number of classes
Zhong and Zhao	2020	Apple	DenseNet121	Centralized	6
Rozaqi and Sunyoto	2020	Potato	Tailored CNN	Centralized	3
Khobragade et al.	2022	Potato	Tailored CNN	Centralized	2
Singla et al.	2022	Rice	Tailored CNN	Centralized	3
Jung et al.	2023	Bell pepper, potato, tomato	Pretrained CNN	Centralized	3
Mac et al.	2024	Tomato	Pretrained CNN	Centralized	10
Prasad et al.	2024	Grape	Pretrained CNN	Centralized	4
Rahman et al.	2024	Tea	Tailored CNN	Centralized	4
Kini et al.	2024	Black pepper	Pretrained CNN	Centralized	6
Vats et al.	2024	Jute	Tailored CNN	FL	4 with FedAvg
Joshi et al.	2024	Sunflower	Tailored CNN	FL	5 with FedAvg
Suryavanshi et al.	2024b	Broccoli	Tailored CNN	FL	5
Upadhyay et al.	2024	Aloe vera	Tailored CNN	FL	4 FedAvg
Hari and Singh	2025	Apple, maize, tomato	Tailored CNN	FL	7
Fahim-UI-Islam et al.	2024	Wheat	Tailored CNN	FL	4 with FedMax
Aggarwal et al.	2024	Rice	Pretrained CNN	FL	4 with FedAvg
Suryavanshi et al.	2024a	Cauliflower	Tailored CNN	FL	5 with FedAvg
Singh et al.	2024	Apple, blueberry, cherry, corn, grape,	ViT	FL	3 FedAvg
Shrivastava et al.	2024	Apple, corn, potato, tomato, grape,	ViT	FL	7
Hari and Singh	2025	Potato, strawberry, tomato, banana,	Tailored CNN	FL	10–13

Among the latest methods for detecting plant diseases, ViTs have shown great potential in centralized systems, owing to their ability to model global relationships through self-attention mechanisms. In addition to enhancing accuracy, the selection of ViT over traditional CNN-based architectures was influenced by its representational benefits and computational considerations. Unlike CNNs that depend on localized receptive fields, ViTs utilize self-attention to capture long-range dependencies. This is particularly advantageous for detecting mango leaf diseases where lesions tend to be small, dispersed, and affected by the overall context. This improves generalization across intricate backgrounds.

From a computational standpoint, ViTs generally necessitate larger training datasets and incur greater memory costs due to the quadratic complexity of attention. These challenges have been addressed through dataset augmentation, meticulous hyperparameter optimization, and GPU-accelerated training, ensuring operational efficiency. When compared to deeper CNNs such as EfficientNet-B7, ViTs provided a superior equilibrium between accuracy and inference speed, as the self-attention mechanism minimizes redundant convolutions. Furthermore, ViTs are inherently scalable; their patch-based input allows for better adaptability to high-resolution images without a proportional increase in computation, unlike CNNs that scale quadratically with resolution.

Therefore, the use of ViT is warranted not only because of its accuracy but also its capacity to efficiently capture global dependencies, scalability to larger input resolutions, and competitive inference performance in comparison to very deep CNNs.

Consequently, the current study suggests the incorporation of ViTs into an FL framework to improve the accuracy, generalization, and privacy of systems designed for detecting plant leaf diseases. By utilizing the attention mechanisms of ViTs within FL, the proposed strategy seeks to:

- Enhance classification performance across diverse client data.
- Safeguard data privacy without compromising model quality.
- Facilitate scalable implementation in resource-limited agricultural contexts.

The primary focus of this study was to characterize diseases in mango plant leaves, while also examining the impact of:

- Device heterogeneity (variations in client devices).
- The number of local iterations per device during federated training.

The role and effectiveness of the ViT attention mechanism in enhancing detection accuracy was assessed.

The contributions of this research are listed here:

1. Explore the impact of attention mechanism for characterized plant leaf disease using FL.
2. Study the impact of varying devices and iteration within the devices in the experiment of leaf disease detection.
3. Explored the advantage of FL in plant leaf disease detection.

3. Materials and methods

This section of the paper covers the details of the dataset used and the methodology applied to characterize leaf disease.

3.1. Dataset

To address the issues of nonindependent and identically distributed (non-IID) data and class imbalance in FL, particularly in the context of agricultural disease detection, researchers have put forward various strategies. This study used a dataset with eight classes, sourced from a prominent open-source repository, to train local clients within an FL framework. The dataset contains images of leaves afflicted by seven prevalent diseases—powdery mildew, anthracnose, dieback, bacterial canker, sooty mould, gall midge, and cutting weevil—alongside images of healthy leaves. Comprehensive statistics regarding the dataset are provided in Table 2, while representative samples are illustrated in Figure 1.

Although the current study is based on a balanced dataset, it is crucial to recognize that actual agricultural data is often imbalanced. Typically, only a limited number of samples represent rare diseases, while more common diseases are disproportionately represented. This imbalance presents several challenges within federated environments. In particular, clients may only have data related to a restricted subset of classes, leading to non-IID data distributions. Such heterogeneity in data poses a significant challenge in FL, as it can result in biased local updates, model bias, and a decrease in the speed of global convergence.

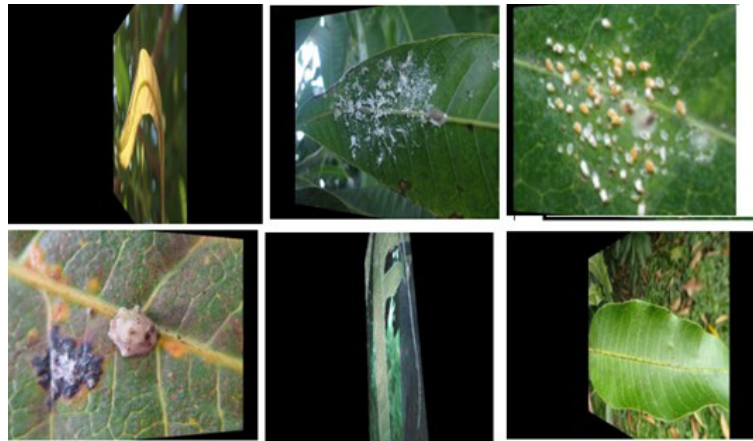
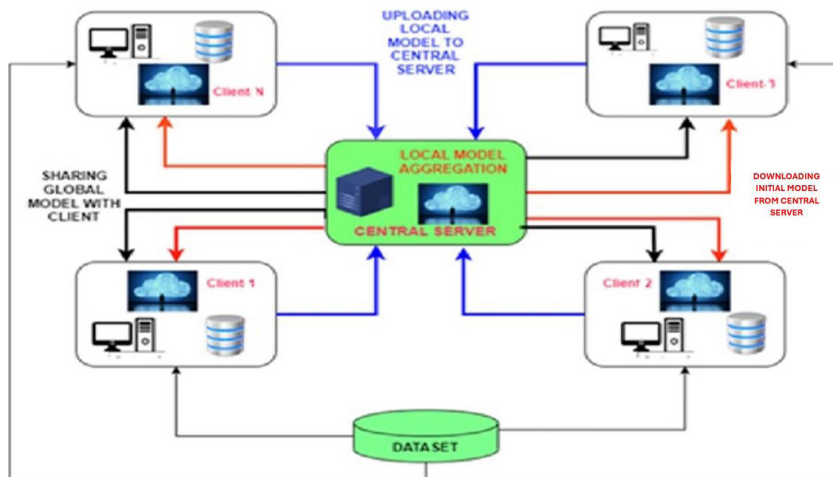
Various operations such as augmentation and data sampling were carried out to address the challenges of non-IID data and class imbalance in FL.

4. Attention mechanism-based deep neural network for FL

In a decentralized approach as FL the multiple clients train their local model and collaborate to build a global model (Li et al., 2020; Hassan and Maji, 2022). Figure 2 illustrates the proposed methodology where the local machines are trained with a publicly available mango leaf disease dataset and afterward collaborate to build or improve a global model. In the network, ViT was utilized to classify plant leaf images. The following section describes the complete FL procedure.

Table 2. Dataset description.

Disease type	Count
Anthracnose	500
Bacterial canker	500
Cutting weevil	500
Dieback	500
Gall midge	500
Powdery mildew	500
Sooty mould	500
Healthy Leaf	500

**Figure 1.** Sample images.**Figure 2.** Proposed methodologies.

4.1. Vision transformer

This section discusses the design of the ViT with complete formal models utilized in FL for classification. The ViT standard design is depicted in Figure 3. Every ViT block

comprises a self-attention layer to estimate attention chances between query vector v and key matrix M and a feed-forward network with two multilayer perceptron layers (Chen et al., 2022). In the network, multihead

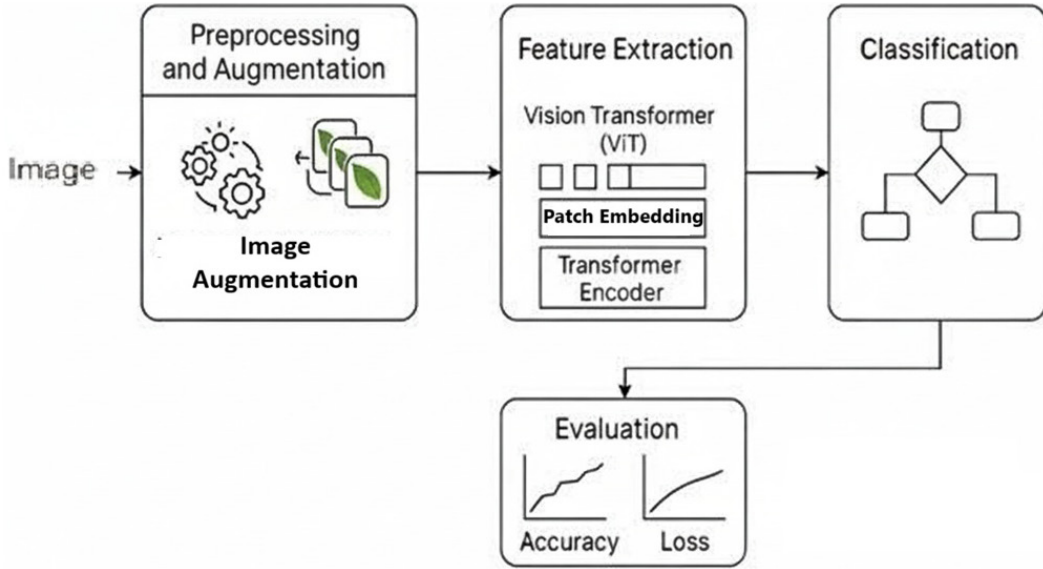


Figure 3. Methodology for detection with ViT.

attention was used to ensure attention mechanism in every part of network. The learning process in the network was ensured with layerNorm (LN), as given in Equations 1–8.

Let X be the input image divided into N patches, each represented as x_i , and $E(x_i)$ be the embedding of patch x_i .

$$E(x_i) = W_{patch(x_i)} \text{ (Eq. 1)}$$

$$PE(x_i) = W_{pos(i)} \text{ (Eq. 2)}$$

$$PE(x_i) = W_{pos(i)} \text{ (Eq. 3)}$$

$$Z_i = LN(E(x_i) + PE(x_i)) \text{ (Eq. 4)}$$

$$A(z) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \text{ (Eq. 5)}$$

$$FFN(Z_i) = Relu(Z_i W_1 + b_1) W_2 + b_2 \text{ (Eq. 6)}$$

$$Z'_i = LN(A(Z_i) + Z_i) \text{ (Eq. 7)}$$

$$Z'_i = LN(FFN(Z_i) + Z_i) \text{ (Eq. 8)}$$

Here the feature map output represented with Equation (9).

$$O_i = softmax(Z'_i W_{class} + b_{class}) \text{ (Eq. 9)}$$

Figure 2 illustrates the comprehensive operational procedure of FL that was carried out through the client and server processes. The complete identification and classification process is presented in Figure 3, which was implemented within a federated environment.

The federated process consisted of multiple steps.

The server machine was initialized using a transformer model that incorporates an attention mechanism.

The second phase commences with the initialization of the client machine, which utilizes an attention mechanism-based transformer along with local data intended for training local models.

The central server will receive updates indicating that the local machine has completed its training process.

Client models are aggregated and subsequently uploaded to the centralized server.

Ultimately, the clients are provided with the enhanced consolidated model produced at the global machine for further assessment under local conditions using the testing dataset.

In this methodology, the entire process is reiterated several times until a suitable model with the correct parameters is achieved. The ultimate model will be distributed among all clients for subsequent actions. The algorithm outlined below is predicated on the functions of two roles: CProcess and SProcess, as detailed in Algorithms 1 and 2, respectively. This methodology facilitates varying levels of client permissions, allowing clients to select their degree of involvement while participating in the FL process. Additionally, it upholds privacy by permitting clients to share model updates or gradients rather than raw data.

Algorithm 1: CProcess; The number of clients is represented with C and the client models are represented with A_L . The batch size and epochs for the client model are represented with B_{ak} and E_i .

1. CProcess(C, A): // client process executes on Client (C)
2. $Ba_k = (\text{Fragment } D_k \text{ into different batches } b_a)$
3. For every local epochs E_i do:
4. For batch $b_a \in B_{ak}$ do
5. $A_{L\beta} A_L - L_r (\text{CPr}(A_L; b_a))$
6. EndFor
7. EndFor
8. Return A_L to Server

Algorithm 2: SProcess: The communication rounds represented with R with C and the client models are represented with A_L .

1. SProcess:
2. Initialize A_{L0}
3. For each communication round $R = 1, 2, \dots$ do
4. $D_R \beta$ Data
5. For each C in D_R process simultaneously do
6. $A_{t+1}^R = \text{CProcess}(k, A_{Lt})$
7. EndFor
8. $A_t = \frac{1}{n} \sum_{R=1}^R n_r * A_{R+1}^R$
9. EndFor

4.2. Sampling-based client selection process (CProcess)

The client process occurs on all client machines. The objective of client process is to define the behavior of all the available clients. The clients initiate with the CProcess to update the global model at the global server (Wang et al., 2022). Several parameters are required for operation, one of which is fixed (the number of experiment iterations) and three of which are variable (the number of clients, learning rate, and batch size). The number of the client is decided with a sampling algorithm that allows dynamic client selection at the time of initialization. Here are the details of the client process:

1. Initialization:

- Define the parameters for the algorithm, such as the number of clients to select in each round (e.g., K clients), the selection criteria, and any hyperparameters.
- Initialize the global model with an initial model or use an existing pretrained model.

2. Client sampling:

- Randomly sample a subset of clients from the available client pool. The number of clients to sample can be fixed (K clients per round) or determined dynamically based on certain criteria.

3. Client ranking:

- Rank the selected clients based on some criteria that assess their suitability for participation in the current round. The criteria can include factors like device performance, available data, or historical performance in previous rounds.
- Common ranking criteria include:

- Local update quality: Evaluate the quality of the updates. The clients can provide by measuring how much their local models deviate from the global model.
- Data diversity: Assess the diversity of data that clients possess. Clients with diverse datasets may be given higher priority.
- Communication cost: Consider the communication cost associated with each client, as clients with high communication costs might be deprioritized.
- Resource availability: Consider the computational resources available on each client, ensuring that they can perform the required computations.

4. Client selection:

- Select the top K clients from the ranked list to participate in the current round. These clients will download the global model, perform local updates, and initiate model updates.

5. Model aggregation:

- The client models are aggregated to generate global models. Common aggregation methods include federated averaging, weighted averaging, or other custom aggregation techniques.

6. Repeat:

- Repeat steps 2–5 with a predefined number of communication rounds or until the convergence criteria are met.
- Optionally, adjust the selection criteria and parameters dynamically.

7. Termination:

- Define stopping criteria after achieving satisfactory results.

8. Final model assessment:

- After the FL process concludes, evaluate the final global model on a held-out validation dataset to assess its performance.

9. Deployment:

- Deploy the final global model for inference or use it for downstream tasks.

These steps provide a high-level overview of a sample-based client selection algorithm for FL.

4.3. Server process (SProcess)

In the FL deep learning approach, the server process is used to aggregate client models. The process is used to select client models from various locations and aggregate them into a global model (Trivedi et al., 2021). The global model will be shared with all the clients for further testing. In this process, the aggregation algorithm is used to generate a composite model. Here are some commonly used aggregation algorithms in federated deep learning:

1. FedAvg: In FedAvg, every participant trains its native model on its data and shares it with the server. The central server computes the weighted average of these

model updates to obtain a global model. FedAvg typically performs well when the local data distribution is relatively balanced across participants.

2. FedSGD: FedSGD is like traditional stochastic gradient descent (SGD) but operates in a federated setting. Each participant computes the gradient of their local loss and sends these gradients to the central server. The server then sums these gradients (e.g., by averaging) to update the global model.
3. Federated averaging with momentum (FedAvgM): FedAvgM extends FedAvg by incorporating momentum into the aggregation process. This helps stabilize the learning process by considering past updates when computing the weighted average of model updates.
4. Federated Adam (FedAdam): FedAdam is an adaptation of the Adam optimization algorithm for FL. It combines the benefits of adaptive learning rates and momentum. Participants compute local gradients and send them to the central server, where a global Adam optimizer aggregates and updates the model.
5. Secure aggregation: In cases where privacy and security are significant concerns, secure aggregation algorithms are used. These algorithms ensure that the model updates are aggregated without showing the individual updates. Secure multiparty computation techniques like homomorphic encryption or cryptographic protocols such as secure aggregation protocol are used to achieve this.
6. Custom aggregation schemes: Depending on the specific FL problem and requirements, custom aggregation schemes can be designed. These may involve weighted aggregation, differential privacy mechanisms, or other domain-specific techniques.

In our case, we used the FedAdam approach. This aggregation procedure is well described in Algorithm 2 (SPProcess).

4.4. Deep learning architecture

In this paper, several pretrained deep neural networks were used for experiments.

- VGG-16: This model is based on stacked architecture. The network is based on a small convolution filter of 3×3 and allows small numbers of parameters (Rumpf et al., 2010). This network consists of 13 convolution layers followed by the ReLU layer and the max pool layer.
- ResNet50: This network compares 50 layers (48 convolutional layers, one max-pooling layer, and one average pool layer). It stacks residual blocks based on the VGG network. A network can be enhanced with a convolution layer and contain a bottleneck block to avoid leakage issues. It has a very basic network design with the same number of filters based on a feature map in each layer of the network.

- DenseNet-121: The architecture of this network is densely connected. The convolution layers are connected through a dense network by stacking dense blocks. The dense blocks are the aggregation inputs from previous layers (Esgario et al., 2020). A variety of networks are based on the structure of dense layers. The DenseNet consists of 7×7 convolution layers, followed by 3×3 convolution layers, etc. Additional dense blocks make it a deep neural network.
- MobileNet-V2: This network is primarily designed for mobile and embedded systems. The network uses depth-separable convolution to design a lightweight network. The network was designed to attain speed and efficiency after reducing the number of parameters. In lightweight MobileNet, each convolution channel produces a corresponding feature map.
- Inception-V3: This is an advanced extended version of the Inception-V1 model. Several improvements such as label smoothing and factorized convolutions have been incorporated into the Inception network. This network comprises 42 layers and has a low error rate.
- Visual transformer (ViT): Transformer architecture-based visual model, originally designed for natural language processing. It is an attention mechanism-based model that makes it different from other deep learning models. This model produces exceptionally better results in the context of image processing.

In this study, various deep learning models listed above were used to implement FL for plant leaf disease detection. The complete experimental setup was established with the listed models. The next section covers the experiments performed with federated architecture.

4.5. Experimental setup

The entire experiment was performed with 3.20 GHz Intel third-generation RT Cores, 400 fourth-generation Tensor Cores, 12,800 CUDA cores, and 32 GB of graphics memory running Fedora 4 LTS 64 bits. PyTorch and Keras Python libraries were used to set up deep learning environments on machines, and a mango plant leaf dataset was used for the experiment. Table 3 shows the optimized parameters used for the complete experiment. Hyperparameter tuning was conducted using a hybrid search strategy, starting with a random search to extensively investigate the parameter space, followed by a grid search in promising areas to fine-tune optimal values. This two-phase method guaranteed both exploration and detailed optimization, resulting in consistent convergence across runs.

4.6. Experiment evaluation metrics

Model performance can be evaluated based on several parameters. The performance parameters were computed on four different values: true positive (T_p), false positive (F_p), true negative (T_N), and false negative (F_N) rates (Bouakkaz et al., 2025).

Table 3. Optimized parameters for experiment setup.

Client	Epoch/client	Batch size	Rounds	Classes	Learning rate
Variable	5, 10	16, 32, 64	10, 30, 50, 100	8	0.01, 0.01, 0.0001

Accuracy: Represents the proportion of the correct prediction among all predictions:

$$Acc = (T_p + T_N) / (T_N + T_p + F_N + F_p)$$

Precision: A value that measures the ability of the model to not predict a negative sample as positive:

$$Pre = T_p / (T_p + F_p)$$

Recall: Represents all positive results generated by the model (Ranjan et al., 2015):

$$Re = T_p / (T_p + F_N)$$

F1-score: The recall value is used to measure the global performance of a model and computed from the harmonic mean of precision and recall. Formally, recall can be expressed with F1 (Prabu and Chelliah, 2022):

$$F1 = 2 * \frac{Re * Pre}{Re + Pre_p}$$

4.7. Experiment result analysis

Several models were tested with variations in the numbers of clients, communication rounds, and number of epoch values using the mango leaf dataset. Model performance was compared on the following aspects:

1. The impact of a variable number of clients
2. The number of iterative communication rounds.
3. The epoch value impact of an experiment.
4. Performance comparison concerning the dataset used for training purposes.

5. Impact of the number of clients on performance

Here, the impact of the variable number of clients was evaluated on decentralized FL during the training process. The experiment was performed with 3, 5, and 7 clients. Other parameters such as communication round and epoch values were fixed for all the scenarios.

Table 4 shows that attention-based ViT outperformed other models for all client numbers and other hypermeter settings. The accuracy and precision were closer to 99.5%. Other pretrained models listed in Table 4 also performed well but their performance was slightly worse than attention-based light weight ViT. The performance of the Inception network for different client configurations was very poor. Table 3 and Figure 4 show that attention-based ViT was better in the case of all evaluation parameters.

Despite the complexity of architecture, ViT performed well. The results show that training costs are reduced with ViT, highlighting another advantage of the network.

Figures 5 and 6 represent the result of federated architecture with three clients. The attention-based lightweight ViT model produced better outcomes and reduced training time and energy consumption.

The MobileNetV2 network was designed for low energy consumption and computation cost for training. The results show that the performance of MobileNetV2 is slightly lower than the ResNet50 for configurations with many clients. However, it is a better choice for configurations with fewer clients.

Even though ViT networks are complex, they can be the best choice if an attention mechanism is needed. Attention mechanisms are essential in the case of leaf disease detection for extracting a region of interest from images. Figures 7 and 8 show the results where the number of clients has increased from 3 to 5. Accuracy, precision, recall, and F1-score were higher for attention-based ViT—outperforming other models with a 98.9% accuracy rate.

6. Impact of the number of communication rounds on performance

The number of rounds was shared with clients and servers. Various numbers of rounds (30 and 50) were considered. The 3-client configuration was used in these tests because the models showed better performance in this configuration. Table 5 shows the result of the experiment performed.

The results presented in Table 5 show that the number of communication rounds has a direct impact on the performance of most models. ViT outperforms other models for all communication round tests, with accuracy, precision, recall, and F1-score consistently above 98%. Figures 9 and 10 show the performance evaluation for all the models with 100 communication rounds with three clients.

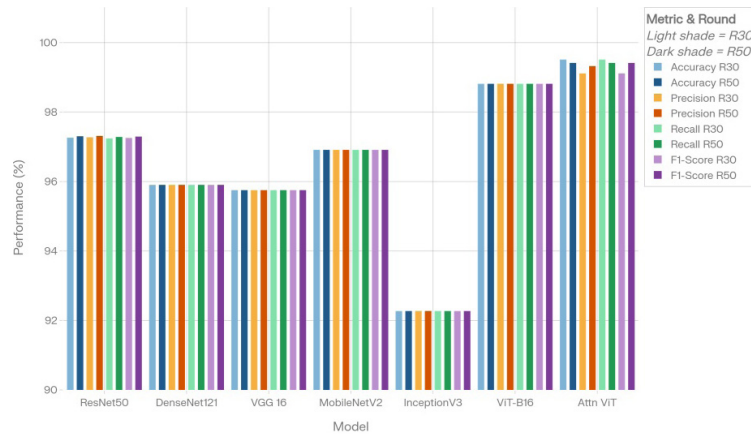
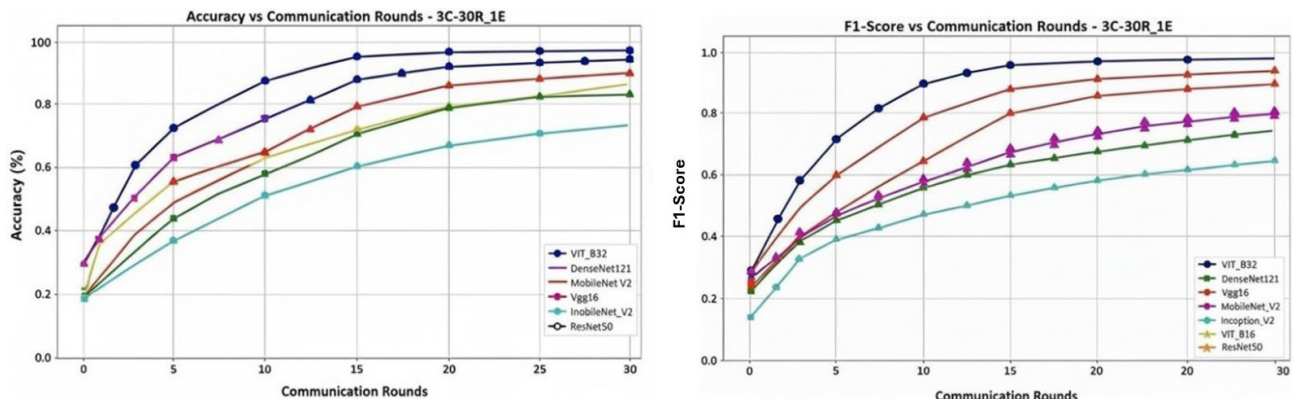
7. Impact of the local iterations

Table 6 shows two epoch values (1 and 5) and in both setups, the performance of ViT models was best. The performance of the other models was slightly lower than the ViT models. The performance analysis of different epoch values is shown Figure 11.

Considering centralized CNN approaches (Das et al., 2025) and decentralized FL approaches with an attention

Table 4. The performance of different models with varying numbers of clients.

	Client = 3				Client = 5				Client = 7			
	Recall	F1-score	Accuracy	Precision	Recall	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall
ResNet50	96.72	96.73	96.74	96.75	96.76	96.77	96.78	96.79	96.80	96.81	96.82	96.83
DenseNet121	97.38	97.38	97.38	97.38	97.38	97.38	97.38	97.38	97.38	97.38	97.38	97.38
VGG16	95.23	95.23	95.23	95.23	95.23	95.23	95.23	95.23	95.23	95.23	95.23	95.23
MobileNetV2	96.39	96.39	96.39	96.39	96.39	96.39	96.39	96.39	96.39	96.39	96.39	96.39
InceptionV3	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75
ViT-B16	98.29	98.29	98.29	98.29	98.29	98.29	98.29	98.29	98.29	98.29	98.29	98.29
Attention-based ViT	98.99	98.59	98.99	98.59	98.89	98.89	98.8	98.89	98.79	98.79	98.79	98.79

**Figure 4.** Comparison of various pretrained models with attention-based ViT.**Figure 5.** Accuracy (left) and F1-score (right) vs communication rounds for three clients.

mechanism (Hari and Singh, 2025) with the mango plant disease dataset, the decentralized approach with FL outperforms centralized CNNs, producing better results while retaining privacy and confidentiality of data.

The results indicate that deep learning architecture performance in the FL approach can be improved after

optimizing the hyperparameters. Both the quality and quantity of the training data affected the obtained results. Differences in the size of client data and number of classes can create variation in the result of the aggregated model. Thus, the adjustment of hyperparameters is required before experiments.

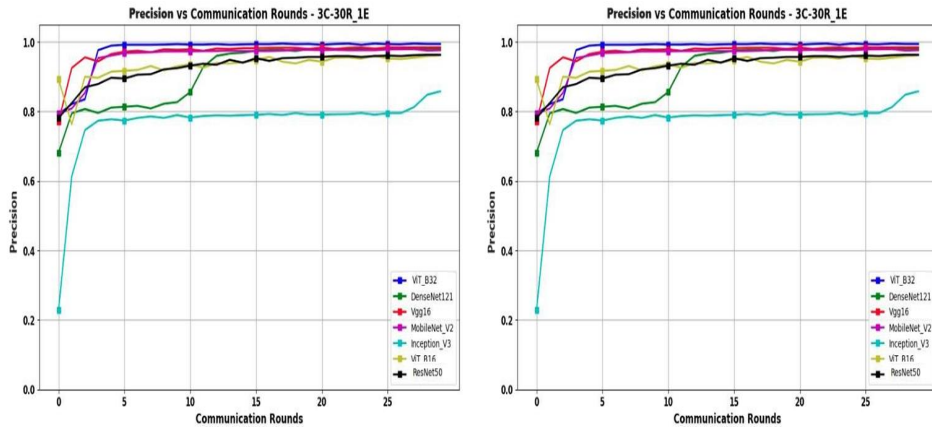


Figure 6. Precision (left) and recall (right) vs communication rounds for three clients.

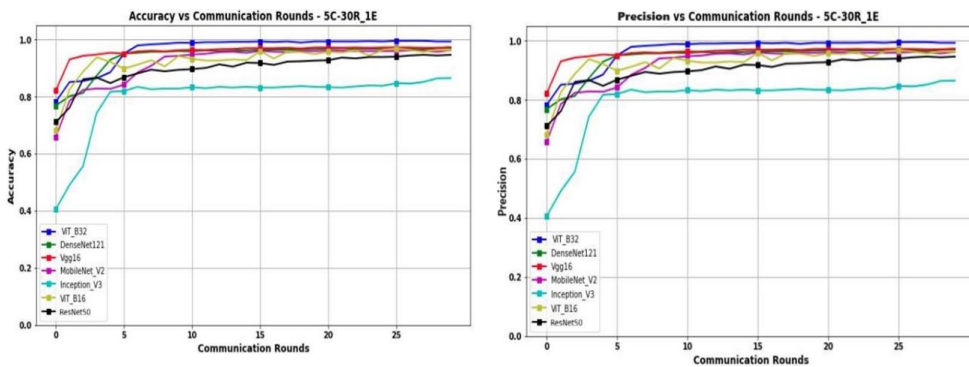


Figure 7. Accuracy (left) and precision (right) vs communication rounds for five clients.

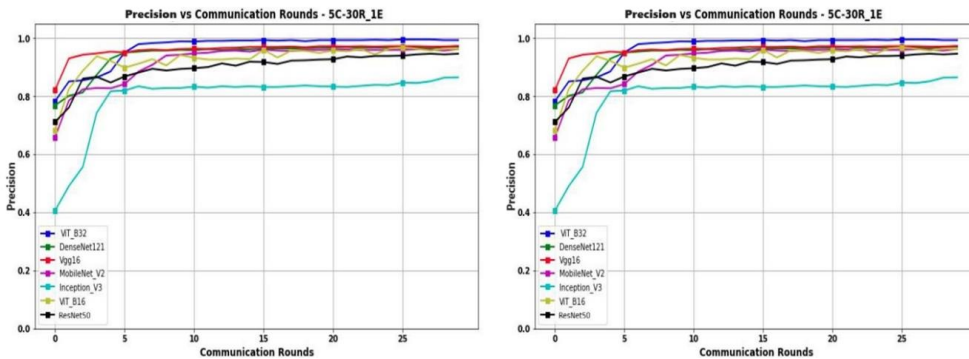


Figure 8. F1-score (left) and recall (right) vs communication rounds for five clients.

The updated confusion matrix in Figure 12 illustrates the performance of the federated ViT model across eight mango leaf disease categories: anthracnose, bacterial canker, cutting weevil, dieback, gall midge, powdery mildew, sooty mould, and healthy leaf.

The multiclass receiver operating characteristic–area under the curve (ROC-AUC) results presented Figure 13 show the ability of the model to discriminate between each disease class. Each curve corresponds to a one vs rest classification, and the AUC scores reflect the predictive

confidence for each class. Higher AUC values indicate better class separation capabilities of the model.

8. Conclusion

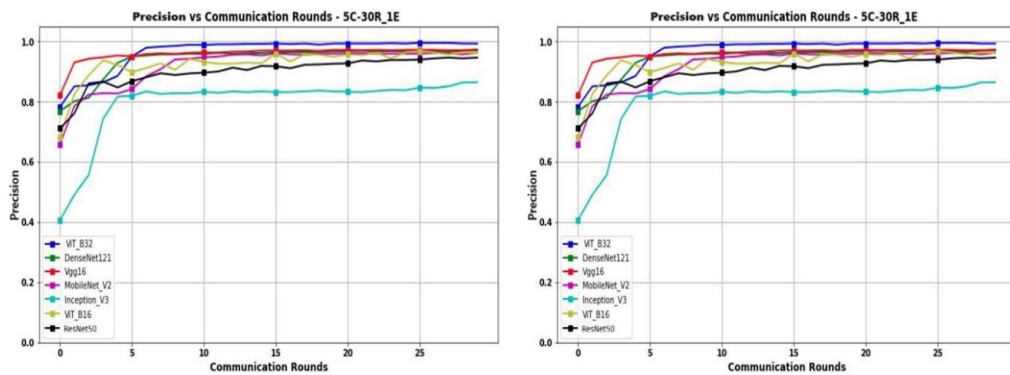
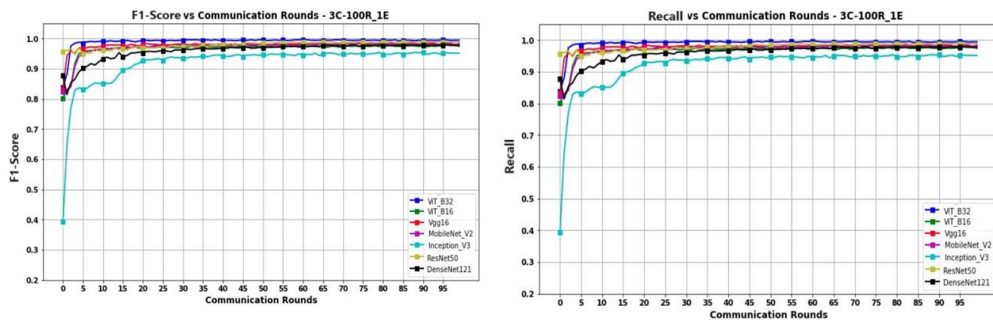
In this study, the decentralized FL method was investigated alongside deep learning architectures to classify plant leaf diseases. Various deep learning architectures were examined to conduct experiments using the mango plant leaf disease dataset. The mango dataset was used to train models of local clients and the results were distributed

Table 5. The impact of the number of communication round impact on performance.

	Round 30				Round 50			
Models	Recall	F1-score	Accuracy	Precision	Recall	Accuracy	Precision	F1-score
ResNet50	96.72	96.73	96.74	96.75	96.76	96.77	96.78	96.79
DenseNet121	95.38	95.38	95.38	95.38	95.38	95.38	95.38	95.38
VGG16	95.23	95.23	95.23	95.23	95.23	95.23	95.23	95.23
MobileNetV2	96.39	96.39	96.39	96.39	96.39	96.39	96.39	96.39
InceptionV3	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75
ViT-B16	98.29	98.29	98.29	98.29	98.29	98.29	98.29	98.29
Attention-based ViT	98.99	98.59	98.99	98.59	98.89	98.89	98.8	98.89

Table 6. Model performance with varying number of epochs.

	Epochs = 1				Epochs = 5			
Models	Recall	F1-score	Accuracy	Precision	Recall	Accuracy	Precision	F1-score
ResNet50	96.72	96.73	96.74	96.75	96.76	96.77	96.78	96.79
DenseNet121	97.38	97.38	97.38	97.38	97.38	97.38	97.38	97.38
VGG16	95.23	95.23	95.23	95.23	95.23	95.23	95.23	95.23
MobileNetV2	96.39	96.39	96.39	96.39	96.39	96.39	96.39	96.39
InceptionV3	91.75	91.75	91.75	91.75	91.75	91.75	91.75	91.75
ViT-B16	98.29	98.29	98.29	98.29	98.29	98.29	98.29	98.29
Attention-based ViT	98.99	98.59	98.99	98.59	98.89	98.89	98.8	98.89

**Figure 9.** Accuracy (left) and precision (right) vs communication rounds for three clients.**Figure 10.** F1-score (left) and recall (right) vs communication rounds for three clients.

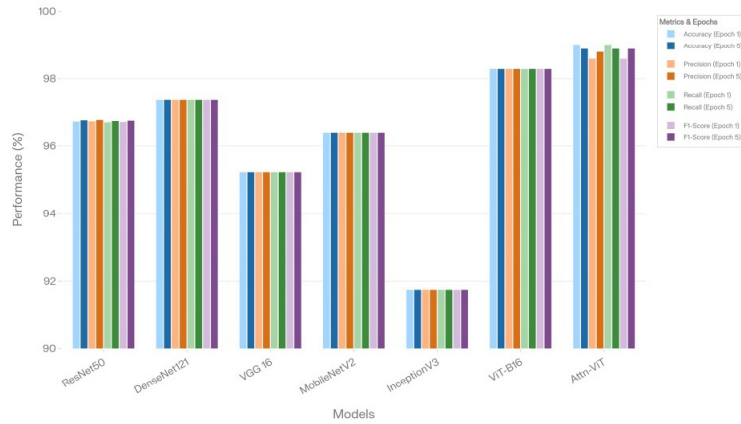


Figure 11. Analysis of client epochs.

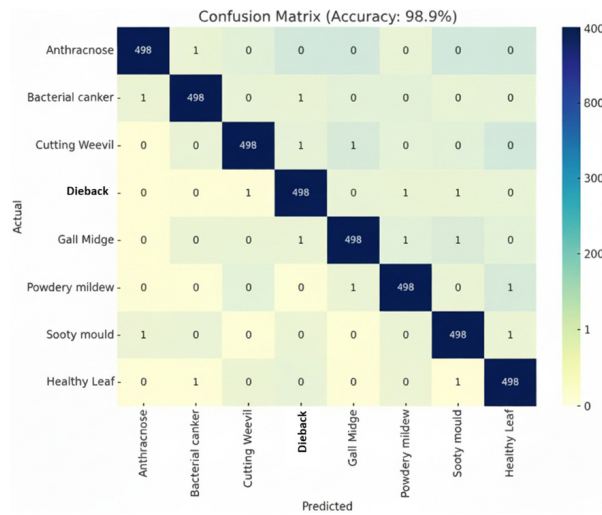


Figure 12. Confusion matrix.

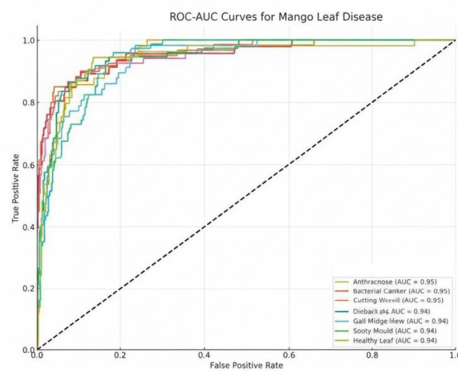


Figure 13. AUC plots.

from a centralized server. Subsequently, the trained model was aggregated using an aggregation algorithm at the centralized server.

This study explored several deep learning architectures, including ResNet50, MobileNetV2, VGG, InceptionNet,

and ViT within an FL framework. The deep learning architectures were fine-tuned with hyperparameters. The entire experiment was conducted with different numbers of clients, communication rounds, and epochs. Regardless of these parameters, the global model utilizing ViT was

superior in performance, consistently reaching accuracy and precision rates of up to 98.9%.

This experiment was conducted using a single plant leaf disease dataset; however, future work should focus on experimenting with a variety of datasets. These findings suggest that the FL approach is vital for classifying plant leaf diseases. In future studies, we will investigate architectures with multiple disease datasets and explore additional facets of FL.

These results show that the FL approach can play a crucial role in plant leaf disease classification. In future research, we will try to explore architecture with multiple disease datasets and explore some other aspects of FL.

Ethical approval and consent to participate

Ethical approval and consent to participate were not required for this study.

Consent for publication

All authors have voluntarily agreed to participate in and consent to the publication of this paper.

Data availability statement

The data are available from the authors upon reasonable request.

Conflict of interest

The authors of this research article declare no conflicts of interest in preparing this research article.

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Author contributions

The authors contributed equally to the preparation of this manuscript.

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